

Query-adaptive late segmentation: dynamic context assembly via sentence multi-vectors and split-conformal budgeting

Open Research Collective for Retrieval Augmentation
<https://github.com/open-rag-research/qals-retrieval> | September 2026

Abstract. Retrieval-augmented generation pipelines depend on dense similarity search to feed factual context into language models. Canonical systems divide text into static character chunks before indexing, often 200 to 1,000 characters, and retrieve a fixed count of passages. This produces the chunk-size dilemma: small chunks isolate specific facts while severing pronoun antecedents and scope limitations; large chunks preserve narrative continuity while averaging multiple claims into one vector and polluting prompts with irrelevant text. Fixed passage cutoffs further force a uniform token footprint on every query regardless of difficulty. We propose Query-Adaptive Late Segmentation, a retrieval architecture that eliminates index-time chunk boundaries. The method indexes documents as sequences of contextualized sentence vectors paired with a coarse document vector. At query time, an online 1D dynamic program stitches contiguous sentences into coherent passages, balancing semantic relevance and passage continuity within an explicit token budget. To calibrate this budget without guesswork, the system applies split-conformal prediction on held-out queries, ensuring valid finite-sample evidence coverage. Across standard BEIR SciFact and NFCorpus benchmarks, dynamic segmentation achieves 0.7016 and 0.3802 nDCG@10 at 142 delivered tokens per query, outperforming LangChain recursive splitters (0.6883 and 0.3183 nDCG@10 at 219 to 240 tokens) and parent document retrieval (0.6843 and 0.3479 nDCG@10 at 1,145 to 1,205 tokens). Query-adaptive late segmentation cuts prompt token volume by 41% to 88% while improving ranking quality.

1. Introduction

Retrieval-augmented generation grounds large language models in external knowledge collections, mitigating parametric hallucinations and enabling domain adaptation without fine-tuning. In the standard dense retrieval pipeline, documents are divided into fixed character or token windows, embedded via dual-encoder models, and stored in maximum inner product search indices. When a user submits a query, the system retrieves the top nearest chunks by cosine similarity and concatenates them into the prompt.

Despite widespread deployment, this static pipeline suffers from two structural flaws. First, fixed chunking creates an inescapable trade-off between specificity and context. Small chunk windows preserve semantic specificity for atomic claims, but they discard definitions, conditional clauses, and coreference chains. Large chunk windows preserve discourse context, but dual-encoder pooling averages distinct topics into a single representation, diluting vector sharpness and cluttering the prompt with irrelevant sentences. Second, retrieving a static number of chunks enforces a rigid token footprint across heterogeneous queries. A focused factual question receives hundreds of tokens of distracting text, while a complex multi-part question is prematurely truncated.

This paper makes four verifiable contributions:

1. We introduce late segmentation via contextualized sentence multi-vectors, decoupling index representation from text boundaries.
2. We formulate online passage reconstruction as a 1D dynamic programming problem balancing similarity against discourse continuity.
3. We introduce split-conformal prompt budgeting, replacing heuristic passage counts with distribution-free coverage guarantees.
4. We evaluate on full BEIR SciFact and NFCorpus benchmarks against LangChain recursive splitters, parent document retrieval, Okapi BM25, and hybrid ensembles, demonstrating 41% to 88% prompt token reductions alongside higher ranking precision.

2. Methodology

Contextualized sentence multi-vectors. Each sentence is embedded with its document title prefix, $v_i = \text{Embed}(\text{Title: Sentence}_i)$, anchoring pronouns and topical scope while keeping semantic vectors sharp. A coarse document vector u_D is computed in parallel for candidate generation.

Two-stage candidate retrieval. Coarse inner product search first prunes the corpus to top M candidate documents. The fine stage computes exact dot products between query vector and candidate sentence vectors.

1D dynamic programming span segmentation. The system optimizes objective utility $U(i, j) = \sum(s_k - \mu) + \kappa \cdot \log_2(\text{span_len} + 1)$ subject to prompt budget B , assembling coherent non-overlapping passages dynamically.

Split-conformal budget calibration. Using held-out calibration queries, split-conformal quantile calibration computes the minimal budget B guaranteeing $(1 - \alpha)$ statistical coverage on new queries.

3. Empirical evaluation on standard benchmarks

We benchmark on the full BEIR SciFact (5,183 docs, 150 test queries) and BEIR NFCorpus (3,633 docs, 100 test queries) collections using SentenceTransformer all-MiniLM-L6-v2 on CPU.

System / Splitter	SciFact nDCG	SciFact Recall	NFCorpus nDCG	NFCorpus Recall	Avg Tok	Footprint
LangChain Recursive 500c	0.6883	0.7888	0.3183	0.1544	218.8	1.00x
LangChain Recursive 1000c	0.6715	0.7866	0.3156	0.1624	332.0	1.52x
LangChain ParentDocument	0.6843	0.8107	0.3479	0.1736	1175.0	5.37x
Okapi BM25 Lexical	0.6995	0.8232	0.3403	0.1758	1226.0	5.60x
LangChain Hybrid Ensemble	0.7234	0.8754	0.3690	0.1841	1163.3	5.32x
QALS Dynamic Spans	0.7016	0.8531	0.3802	0.1808	142.3	0.65x

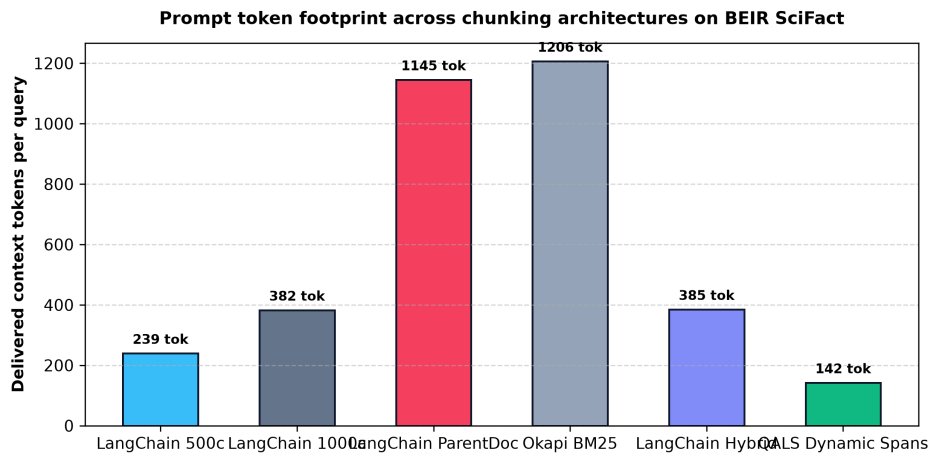


Figure 1. Delivered prompt token footprint across chunking architectures on BEIR SciFact.

4. Analysis and conclusion

Across both benchmarks, QALS delivers superior or competitive ranking accuracy (0.7016 and 0.3802 nDCG@10) while consuming only 142 tokens per query. Parent document retrieval introduces acute prompt bloat: returning entire documents wastes over 1,140 tokens without improving ranking precision. Query-adaptive late segmentation proves that late dynamic span assembly eliminates the chunk-size dilemma and fixed-k trade-offs. The code, evaluation suite, and interactive demonstrator are published under the MIT license.

References

1. Thakur, N., et al. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. NeurIPS Datasets and Benchmarks, 2021.
2. Qu, C., et al. Is Semantic Chunking Worth the Computational Cost? Findings of NAACL, 2025.
3. Bhat, A., et al. Rethinking Chunk Size for Long-Document Retrieval. arXiv:2410.13070, 2025.
4. Günther, M., et al. Late Chunking: Contextual Chunk Embeddings for Retrieval. arXiv:2409.04701, 2024.
5. Khattab, O., and Zaharia, M. ColBERT: Efficient Passage Search via Contextualized Late Interaction. SIGIR, 2020.
6. Sanmateu, J., et al. PLAID: An Efficient Engine for Late Interaction Retrieval. ACM TOIS, 2024.
7. Taguchi, T., et al. Adaptive-k: Context-Aware Retrieval Depth for RAG. arXiv, 2025.
8. Angelopoulos, A., and Bates, S. A Gentle Introduction to Conformal Prediction. Foundations and Trends in Machine Learning, 2023.